



Best smellers

Male moths have some of the best chemical senses in the animal world. Males smell with their feathery antennae, which are specially “tuned in” to scent signals, or pheromones, produced by females that are ready to mate. Using its antennae, this male atlas moth can detect a female up to 6½ miles (11 km) away if it is upwind. The moth’s antenna can sense minute changes in the strength of a pheromone trail. By zigzagging through the trail, the enormous moth—the world’s biggest—eventually makes its way to the female.

Taste buds

With its rough surface, the tongue of an Arctic fox is just the right shape for tasting and manipulating food. Its roughness comes from small bumps called papillae. Some of these bumps have taste buds set in their surface. Shaped like microscopic barrels, taste buds have a collection of sensory hairs that react to food. Each taste bud senses a basic flavor, such as sour, salty, bitter, or sweet. The brain then puts them together to sense taste.

Large numbers of taste buds line trenches within papilla



Magnified view of rounded papilla from back of tongue

